

Formalization of a Dynamic Inhibitory Mask

Architectural draft of an inhibitory layer for transformers
based on tensor projections and precision-weighting in the active inference
framework

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Version 0.5 — May 2, 2026 (added: alternative adaptation schemes for D_{baseline} in §5.2, “collapse” failure mode in §9.5, multi-anchor / subspace $\mathcal{G}$ extension in §9.6)

1 Basic notation

We introduce the following objects:

- $\mathbf{G} \in \mathbb{R}^d$ — **goal anchor vector** in the model’s latent space, normalized: $\|\mathbf{G}\| = 1$.
- $\mathbf{A}_t^{(\ell)} \in \mathbb{R}^{N \times d}$ — **activation tensor** at layer ℓ at time t , where N is the number of tokens in context and d is the hidden state dimension.
- $\mathbf{a}_{t,i}^{(\ell)} := \mathbf{A}_t^{(\ell)}[i, :] \in \mathbb{R}^d$ — activation of token i at time t as a vector.
- $\sigma_i(t) := \frac{\mathbf{a}_{t,i}^{(\ell)} \cdot \mathbf{G}}{\|\mathbf{a}_{t,i}^{(\ell)}\|} \in [-1, 1]$ — **cosine similarity** of token i ’s activation with the goal (drift angle).

The aim of the formalism is to construct a mask $\mathbf{M}_t \in [0, 1]^N$ that modulates each token’s contribution to the main model’s attention.

Notational convention. Where the time t is clear from context, the time index is dropped: σ_i instead of $\sigma_i(t)$, $\mathbf{a}_i^{(\ell)}$ instead of $\mathbf{a}_{t,i}^{(\ell)}$. Full notation is used where temporal dynamics is essential (§5).

2 Basic mask

§2.1–§2.3 present three variants of the function $f : \sigma_i \mapsto M_i$ mapping cosine similarity to a mask weight. §3 gives a fourth variant with a geometric interpretation of an admissibility

corridor. **An implementation chooses one of the four** depending on requirements for interpretability, differentiability, and the desired shape of the transition zone. The parameter τ is present in all variants in the role of “hardness” (the precise semantics differs between variants); the homeostat in §5.2 updates τ for whichever mask variant has been chosen.

2.1 Hard mask (binary)

$$M_i = \mathbb{1}[\sigma_i \geq \theta] \quad (1)$$

A simple binary mask with threshold θ . Non-differentiable — requires a straight-through estimator under end-to-end training.

2.2 Soft sigmoidal mask

$$M_i = \text{sigmoid}(\tau \cdot (\sigma_i - \theta)) \quad (2)$$

- θ — transition point (typically $\theta = 0$, corresponding to “orthogonality to the goal”);
- $\tau \geq 0$ — **hardness** parameter (precision-analog).

Limiting cases: $\tau \rightarrow \infty$ recovers the hard mask; $\tau \rightarrow 0$ gives a uniform $M_i \approx 0.5$.

2.3 Sparse mask via α -entmax

$$\mathbf{M} = \alpha\text{-entmax}(\tau \cdot \boldsymbol{\sigma}) \quad (3)$$

Yields **exact zeros** for irrelevant tokens while preserving differentiability. Equivalent to softmax with controllable sparsity [7].

Normalization remark. α -entmax produces a distribution whose elements sum to 1; for use as an element-wise multiplicative mask (rather than as a distribution), rescaling is needed — for example, $\mathbf{M} = N \cdot \alpha\text{-entmax}(\tau \cdot \boldsymbol{\sigma})$, so that the average element is on the order of 1.

3 Admissibility corridor (safe manifold)

To avoid suppressing useful variation in activations, around the goal vector $\mathbf{G}$ we define an admissibility cone with half-angle α_{safe} . Inside the cone the mask does not interfere; outside it falls smoothly toward zero.

Let $\sigma_{\text{safe}} = \cos(\alpha_{\text{safe}})$. Then:

$$M_i = \begin{cases} 1, & \sigma_i \geq \sigma_{\text{safe}} \\ \left(\frac{\sigma_i}{\sigma_{\text{safe}}}\right)^\tau, & 0 \leq \sigma_i < \sigma_{\text{safe}} \\ 0, & \sigma_i < 0 \end{cases} \quad (4)$$

Geometric interpretation:

- Inside the cone α_{safe} around $\mathbf{G}$ — **free-variation zone**, the mask is inactive ($M_i = 1$).
- In the transition zone $\sigma_i \in [0, \sigma_{\text{safe}})$ — **smooth suppression** with rate controlled by τ .
- For $\sigma_i < 0$ (activation orthogonal or opposite to the goal) — **hard zeroing**.

The parameter α_{safe} sets the **width of the admissibility corridor** — a balance between creativity preservation and protection from drift. In practice it is selected through ablation on failure scenarios.

3.1 Adaptive percentile threshold

In high-dimensional spaces ($d \sim 10^3\text{--}10^4$) random unit vectors concentrate around orthogonality (concentration of measure): for $\mathbf{u}, \mathbf{v}$ uniformly distributed on $\mathbb{S}^{d-1}$, the inner product $\sigma = \mathbf{u} \cdot \mathbf{v}$ has mean 0 and variance $1/d$; as $d \rightarrow \infty$ the distribution asymptotically approaches $\mathcal{N}(0, 1/d)$. Transformer activations are not random (they lie on a low-dimensional manifold of intrinsic dimension on the order of tens), but their distribution can still concentrate in a narrow band of σ_i values, and a fixed threshold σ_{safe} may turn out either too strict or too loose depending on the model state.

An alternative is an **adaptive threshold**, defined as a percentile of the current distribution of similarities:

$$\sigma_{\text{safe}}^{(\text{adapt})}(t) = \text{Percentile}_p(\{\sigma_i(t)\}_{i=1}^N) \quad (5)$$

where $p \in [50, 90]$ is the target percentile (e.g., $p = 75$ means “the mask does not interfere for the top 25% of tokens most aligned with the goal”).

Advantages: automatic normalization to the model state; robustness to scale shifts in activations across layers.

Disadvantages: dependence on the batch/context; in adversarially constructed scenarios (where **all** context tokens are poorly aligned with the goal) the percentile may not cut off anything.

In practice both approaches (fixed σ_{safe} and adaptive $\sigma_{\text{safe}}^{(\text{adapt})}$) are combinable: $\sigma_{\text{safe}}^{(\text{combined})} = \max(\sigma_{\text{safe}}^{(\text{const})}, \sigma_{\text{safe}}^{(\text{adapt})})$.

The choice of $\max(\cdot)$ (rather than $\min(\cdot)$) yields a **more conservative** rule: a token enters the free-variation zone only if it satisfies **both** conditions simultaneously — exceeds both the absolute threshold and the current batch percentile. This aligns with the safety-oriented logic of the formalism: when in doubt, better to inhibit than to pass through. The alternative $\min(\cdot)$ would yield a more liberal rule (one condition suffices) and suits scenarios where the priority is preserving creativity.

4 Mask application

4.1 Direct application to activations (Hadamard)

$$\tilde{\mathbf{A}}_t^{(\ell)} = \text{diag}(\mathbf{M}_t) \cdot \mathbf{A}_t^{(\ell)} \quad (6)$$

That is, each row i of the activation matrix is multiplied by the scalar M_i . This is equivalent to a Hadamard product with broadcasting of the mask across all d columns; the diag form is more formally precise.

4.2 Application via attention masking (preferred variant)

Standard self-attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \quad (7)$$

The modification — additive mask in the logits before softmax:

$$\boxed{\text{Attention}_M(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + \log \mathbf{M}_t^\top\right) V} \quad (8)$$

When $M_j = 0$ for token j , $\log(0) = -\infty$, which after softmax gives an exact zero contribution of token j to the attention output. This is compatible with the standard additive masking mechanism in transformers — **no need to rewrite the model core**.

5 Dynamic homeostat: τ as a function of dispersion

The mask’s hardness should not be static. As context accumulates, activation heterogeneity grows; the homeostat should respond by strengthening suppression.

5.1 Variance of context alignment

As a sensor of heterogeneity we use the **variance** of cosine similarities:

$$D_\sigma(t) = \text{Var}_i(\sigma_i(t)) = \frac{1}{N} \sum_{i=1}^N (\sigma_i(t) - \bar{\sigma}(t))^2, \quad \bar{\sigma}(t) = \frac{1}{N} \sum_{i=1}^N \sigma_i(t) \quad (9)$$

- When all σ_i are close to one another (context homogeneously aligned with the goal **or** homogeneously misaligned) — D_σ is low.
- When σ_i are scattered (some tokens aligned with the goal, some competing) — D_σ is high.

5.2 Feedback rule

$$\tau_t = \min(\tau_{\max}, \tau_0 \cdot \exp(\gamma \cdot (D_\sigma(t) - D_{\text{baseline}}))) \quad (10)$$

where:

- τ_0 — base hardness;
- D_{baseline} — target variance (e.g., exponential moving average of the first K steps of a session);
- $\gamma > 0$ — homeostat gain coefficient;
- $\tau_{\max}$ — upper bound preventing exponential blowup of τ in edge cases (typical value 50–100).

Interpretation. When context becomes heterogeneous (variance of alignments rises), τ rises, the mask hardens, suppression strengthens. This is a functional analog of **tonic dopamine** in the PFC attractor model — stabilizing system dynamics under rising noise.

Lower bound. When $D_\sigma(t) \ll D_{\text{baseline}}$, τ approaches zero and the mask becomes inactive — uniform pass-through. This is correct behavior for homogeneously goal-aligned context. In adversarial scenarios “homogeneous context” can mean “homogeneously hostile” (a jailbreak with consistent offset of all tokens from the goal). A lower bound $\tau_{\min} > 0$ guarantees a minimum level of inhibition:

$$\tau_t = \text{clip}(\tau_0 \cdot \exp(\gamma \cdot (D_\sigma(t) - D_{\text{baseline}})), \tau_{\min}, \tau_{\max}) \quad (11)$$

Typical value $\tau_{\min} \approx 0.5\text{--}1.0$.

Adaptation-scheme remark for D_{baseline} . The target variance can be updated by two schemes with different trade-offs:

- **Exponential moving average (EMA):** $D_{\text{baseline}}^{(t+1)} = (1 - \alpha) \cdot D_{\text{baseline}}^{(t)} + \alpha \cdot D_{\sigma}(t)$. Smooth adaptation, all past influences with decaying weight, simple to implement. Suitable for sessions evolving gradually.
- **Sliding window:** $D_{\text{baseline}}^{(t)} = \text{mean}(D_{\sigma}(t - W), \dots, D_{\sigma}(t - 1))$ — average over the last W steps. Hard cutoff, past beyond the window has no influence. Suitable for sessions with sharp phase transitions, where EMA may “stick” in an outdated baseline.

For agentic scenarios with possible regime shifts, sliding window with $W \sim 50\text{--}200$ steps is recommended.

5.3 Alternative via free energy minimization

A variational alternative to the heuristic in §5.2 is to interpret τ as the **precision** π over next-token policy and update it by minimizing expected free energy. Full derivation requires choices of generative model, q_{τ} , and target distribution p^* aligned with $\mathbf{G}$, and is left to future work.

6 Connection to precision-weighting in active inference

The mask $\mathbf{M}_t$ is functionally equivalent to **precision** π_i for each token i :

$$\pi_i \propto M_i \tag{12}$$

— tokens with high M_i have high precision (the model “trusts” them and passes them on); with low M_i — low (effectively ignored).

In the cortex this function is performed by GABA-ergic lateral inhibition under top-down control of the prefrontal cortex [3, 6]: PFC sends a signal that raises gain on goal-relevant neurons and lowers it on competitors.

7 Architectural draft of the layer

Input context $\mathbf{C}_t$ (tokens 1..N)

```

|
+----> Main transformer (frozen weights)
|       For each attention layer L in {1, ..., L}:
|           Q, K, V <- projections from prev activations
|           logits = QK^T / sqrt(d_k)
```

Biological component	Formal analog
Goal representation in PFC (delay-period activity)	$\mathbf{G}$ — anchor vector
GABA-ergic inhibitory signal	$\mathbf{M}_t$ — mask
Tonic dopamine (gain modulation)	τ_t — dynamic hardness
Conflict monitoring in ACC	$D_\sigma(t)$ — heterogeneity signal that τ responds to

Table 1: Correspondence between biological components and formal objects (functional analogies, not claims of neural realism).

```

|           logits_masked = logits + log(M_t)
|           weights = softmax(logits_masked)
|           output = weights @ V
|
+----> Supervisor module (small, separate, low-D)
|
|   +-- Goal anchor G (frozen at session start)
|   |
|   +-- Compute sigma_i(t) = (a_{t,i} . G) / ||a_{t,i}||
|   |   for all i
|   |
|   +-- Compute D_sigma(t) = Var_i(sigma_i(t))
|   |   -- alignment dispersion
|   |
|   +-- Update tau_t <- min(tau_max,
|   |   tau_0 * exp(gamma * (D_sigma - D_baseline)))
|   |
|   +-- Compute M_t <- f(sigma, tau_t, sigma_safe)
|   |
+----> Broadcast M_t to all attention layers

```

8 Parameters: summary table

All parameters are subject to calibration via ablation on failure scenarios (Mythos-style drift, sycophancy, jailbreaks via context-shift).

9 Directions of development

The formalization specifies the functional mechanism. The following directions are developed in companion work:

1. **Source of $\mathbf{G}$.** $\mathbf{G}$ is extracted from initial context or the model’s long-term priors

Parameter	Role	Typical range
d	Hidden state dimension	768–4096 (model-dependent)
$\mathbf{G}$	Goal anchor vector	Unit vector in $\mathbb{R}^d$
$\sigma_{\text{safe}} = \cos(\alpha_{\text{safe}})$	Admissibility corridor boundary	0.3–0.7
θ	Mask switching threshold	0.0 (full orthogonality)
τ_0	Base hardness	1–10
γ	Homeostat gain	0.1–1.0
D_{baseline}	Target alignment variance	EMA of first K steps
τ_{max}	Upper hardness bound (clipping)	50–100
τ_{min}	Lower hardness bound (optional)	0.5–1.0
p	Percentile for adaptive threshold	50–90

Table 2: Parameters of the formalism with typical ranges.

and must be protected from subsequent context drift and from prompt injection (an injection at session start can poison the goal). Architectural candidates:

- Goal induction phase at session start with $\mathbf{G}$ frozen;
- Multi-timescale architecture with a slow layer [8];
- A separate aggregator module synthesizing $\mathbf{G}$ from multiple sources by analogy with biological convergence of HPC + amygdala + OFC + DLPFC onto PFC;
- **Contrastive encoder**, trained on pairs of goal formulations (positive: “send the email” $\leftrightarrow$ “transmit the electronic message”; negative: “send the email” $\leftrightarrow$ “publish the data”). The encoder maps a goal formulation to a stable point in latent space, invariant to paraphrases and robust to surface-level injections. The approach requires a goal-invariance dataset; the domain of applicability is determined by the training distribution.

2. **Parameter calibration.** The parameters $(\theta, \sigma_{\text{safe}}, \tau_0, \gamma, D_{\text{baseline}}, \tau_{\text{max}})$ are calibrated by a trainable supervisor module via bootstrap end-to-end loss with regularization against collapse, or by manual tuning on benchmarks.
3. **Layers of application.** Mask placement (all layers / top layers / specific subset) is a design choice determined by ablation studies. The biological analogy — PFC modulates sensory input hierarchically — suggests a hierarchical schedule as the first hypothesis to test.
4. **Heterogeneity meta-monitor.** When $D_\sigma(t)$ remains high over T consecutive steps (despite tightening of τ), the goal anchor itself has lost relevance — the scene has changed enough that no part of the context is now aligned with $\mathbf{G}$. The biological analog is an ACC signal of persistent conflict, calling not for local correction but for revision of the goal. System responses: (a) request goal recomputation from

the supervisor; (b) signal to an external orchestrator; (c) halt the session with notification. Implemented as a separate subsystem — extension of the base formalism. Biological parallel: dorsomedial PFC and ACC perform exactly this function of detecting persistent prediction error and initiating strategic switching.

5. **“Collapse” failure mode.** Under aggressive calibration (high γ , low σ_{safe} , no τ_{min} or excessively high τ_{max}) the mask suppresses almost the entire context except a narrow “echo” of the goal $\mathbf{G}$. The model degrades to repeating the goal’s wording — content variety is lost. Observable markers: sharp drop in perplexity *together with* drop in content diversity; repetition of keywords from the goal formulation; reduction in informativeness while topic adherence is formally preserved. **Ablation studies (§10) track this regime** through diversity metrics (distinct-n, self-BLEU) in parallel with drift and perplexity.
6. **Multi-anchor / subspace $\mathcal{G}$.** The basic formalization uses a single unit vector $\mathbf{G}$. For multi-faceted goals with several simultaneously relevant aspects (e.g., “scientific article” = “precise” + “structured” + “data-grounded”), a subspace $\mathcal{G} = \text{span}(\mathbf{g}_1, \dots, \mathbf{g}_K) \subset \mathbb{R}^d$ is used. Cosine similarity is replaced by the normalized projection of the activation:

$$\sigma_i = \frac{\|\text{Proj}_{\mathcal{G}}(\mathbf{a}_{t,i}^{(\ell)})\|}{\|\mathbf{a}_{t,i}^{(\ell)}\|} \quad (13)$$

where $\text{Proj}_{\mathcal{G}}$ is the orthogonal projection onto $\mathcal{G}$. *Note:* this generalization yields $\sigma_i \in [0, 1]$ — the **magnitude** of subspace alignment, with no sign. For mask variants whose semantics depend on a signed σ (such as the $\sigma < 0$ branch in §3), that branch becomes vacuous; a signed extension would require an oriented half-space rather than a subspace. This removes the “slogan-degradation” risk (item 5) for complex goals. Extension parameters: K , training scheme for basis vectors, facet weighting (equal or context-dependent). Natural generalization of the single-vector case, developed in a separate work.

10 Minimal program for empirical validation

To confirm the formalism’s workability in its simplest form:

1. **Baseline measurement** on an available open-source LLM (Llama-3-8B): logit shift when actionable context is added.
2. **Fixed-G prototype:** implement a supervisor with a frozen $\mathbf{G}$ from system prompt, mask applied to the top 1–2 attention layers, parameters tuned manually. Compare logit distributions to baseline.

3. **Adaptive- τ extension:** enable the homeostat over variance and check whether τ responds in the expected way to jailbreak-style context.
4. **Ablation over α_{safe} :** check the trade-off between creativity and drift protection.

Each step is on the order of 1–2 weeks of work on a single GPU.

References

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